

IT Help Desk & Ticketing Management System

API Documentation

Backend: PHP Laravel | Auth: JWT (tymon/jwt-auth) | Base path: /api

All endpoints below are prefixed with the API base URL (e.g. `http://127.0.0.1:8000/api` locally). Endpoints marked “Public” require no authentication. All other endpoints require a valid JWT, sent as “Authorization: Bearer <token>”, obtained from the `/login` endpoint.

1. Authentication

Method	Endpoint	Description	Access
POST	/register	Create a new user account. No role is assigned at registration — an Admin must assign one afterward. Triggers a “New User Registered” notification to Admins.	Public
POST	/login	Authenticate and receive a JWT access token.	Public
GET	/me	Return the currently authenticated user's profile and role.	Authenticated
POST	/logout	Invalidate the current JWT.	Authenticated

2. Tickets

Method	Endpoint	Description	Access
GET	/tickets	List tickets. Results are scoped by role and default to the last 2 months (override with <code>show_all</code> , <code>from</code> , <code>to</code> query params). Supports filtering by <code>category_id</code> , <code>priority_id</code> , <code>status_id</code> , <code>search</code> .	Authenticated (scoped by role)
POST	/tickets	Create a new ticket. <code>reference_no</code> is auto-generated (TCK- <code>{year}</code> - <code>{sequence}</code>).	Employee, Admin
GET	/tickets/{id}	View full details of a single ticket.	Authenticated (scoped by role)
PUT	/tickets/{id}	Update a ticket (status, category, priority, etc.). Status changes are logged to <code>ticket_status_history</code> .	Role-dependent — see Role & Permission notes
DELETE	/tickets/{id}	Delete a ticket.	Employee (own, while Open), Admin
POST	/tickets/{id}/assign	Assign a ticket to an Agent. Logged to <code>activity_logs</code> .	Agent (self-assign only), Admin (assign to any Agent)

Method	Endpoint	Description	Access
GET	/tickets/{id}/activity	Merged, chronological timeline combining activity_logs and ticket_status_history.	Authenticated (scoped by role)
POST	/tickets/{id}/request-cancellation	Request cancellation of an assigned ticket, with a required reason (logged to activity_logs).	Agent (assigned Agent only)
POST	/tickets/{id}/resolve-cancellation	Confirm the cancellation or reassign to a different Agent. Reverts to prior status if reassigned.	Manager

Role & Permission notes: Employees may edit/delete only their own tickets while status is Open. Agents may change status freely but cannot reassign; a Closed ticket is locked from edits for Agents (only Admin can reopen via status change) and locked from reassignment for everyone, including Admin. Managers have view/monitor access only — they cannot assign, edit, or delete tickets.

3. Ticket Comments

Method	Endpoint	Description	Access
GET	/tickets/{id}/comments	List comments on a ticket. Internal (is_internal) comments are hidden from Employees.	Authenticated (scoped by role)
POST	/tickets/{id}/comments	Add a comment to a ticket. Optionally attach an image in the same request. is_internal (team-only note) is Agent-only.	Authenticated
DELETE	/comments/{id}	Delete a comment. Removes any attached file from disk as well.	Comment author, Admin

4. Attachments

Method	Endpoint	Description	Access
POST	/tickets/{id}/attachments	Upload a file attachment to a ticket. Allowed extensions: jpg, jpeg, png, gif, pdf, doc, docx, xls, xlsx, txt, zip. Size limit: 1–10MB. Only filename/path/MIME/size are stored in the database — files live on disk.	Employee (own ticket), Admin

5. Notifications

Method	Endpoint	Description	Access
GET	/notifications	List notifications for the current user.	Authenticated
GET	/notifications/unread-count	Get the count of unread notifications (used for the bell icon badge).	Authenticated

Method	Endpoint	Description	Access
POST	/notifications/{id}/read	Toggle a notification's read/unread state.	Authenticated

Notifications are triggered automatically on: new ticket creation (Managers + Admins notified), assignment (assignee notified), ticket closure (Employee + Managers + Admins notified), and new user registration (Admins notified).

6. Dashboard & Reports

Method	Endpoint	Description	Access
GET	/dashboard/summary	Role-scoped ticket counts by status for the landing dashboard. Agents/Admins also see unassigned tickets.	Authenticated (scoped by role)
GET	/dashboard/report	Ticket counts by status/category/priority and average time-to-resolve. Supports range=month year custom (with start_date/end_date).	Admin
GET	/dashboard/report/export/pdf	Export the current report selection as a PDF (barryvdh/laravel-dompdf).	Admin
GET	/dashboard/report/export/excel	Export the current report selection as CSV (streamed via PHP fputcsv).	Admin

7. AI Features

Method	Endpoint	Description	Access
POST	/ai/suggest-category-priority	Given a ticket title/description, suggests a category and priority from real DB values (never a hallucinated one). Degrades gracefully if no OpenAI key is configured.	Authenticated
POST	/ai/chat	Single-turn Q&A assistant scoped to IT support topics.	Authenticated

8. User Management

Method	Endpoint	Description	Access
GET	/users	List all users and their roles.	Admin
POST	/users/{id}/assign-role	Assign or change a user's role. Required after self-registration, since registration no longer accepts a client-supplied role.	Admin
DELETE	/users/{id}	Delete a user. Blocked if the user has any tickets attached (as submitter or assigned agent), to avoid orphaned references.	Admin

9. Lookup Data

Method	Endpoint	Description	Access
GET	/categories	List all ticket categories (Hardware, Software, Network, Email, Access Request, Other).	Authenticated
GET	/priorities	List all ticket priorities (Low, Medium, High, Critical).	Authenticated
GET	/statuses	List all ticket statuses.	Authenticated
GET	/agents	List all users with the IT Support Agent role, for assignment dropdowns.	Authenticated
GET	/roles	List all available roles.	Authenticated
